

Development of an integrated sensory-emotional-neural response profiling method for horticultural tactile stimulation based in psychological state

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1. Abstract

Tactile stimulation with horticultural materials may promote psychological well-being, but the neurophysiological basis of group-specific emotional responses remains unclear. This pilot exploratory study examined whether real and artificial horticultural tactile stimuli differentially evoke positive emotions according to perceived stress and sleep quality. Thirty-three adults explored 24 tactile stimuli, including 12 real vegetable materials and 12 artificial counterparts, and rated their positive emotional responses on a 7-point Likert scale. Functional near-infrared spectroscopy (fNIRS) responses were analyzed as channel-group profiles across four psychological state groups: Low_PSS, Mid_PSS, Good_PSQI, and Bad_PSQI. Multivariate analyses, including partial least squares discriminant analysis (PLS-DA), principal component analysis (PCA), correspondence analysis (CA), and hierarchical cluster analysis (HCA), were applied to identify group-related sensory and neural patterns. Real vegetable materials elicited higher positive emotion scores than artificial materials across tactile conditions ($p < 0.05$). PLS-DA suggested stress- and sleep-related response patterns, with Cabbage_R and Chili_R contributing most strongly to stress-related discrimination (VIP = 1.40), whereas Tomato_R showed the highest contribution to the sleep-related axis (VIP = 1.65). PCA and CA supported a shared positive-response structure in which CH12, approximating the left DLPFC, and CH15, approximating the left IFG/VLPFC-lateral OFC region, co-varied with key sensory variables. These preliminary findings suggest that perceived stress and sleep quality may independently modulate positive emotional and prefrontal responses to horticultural tactile stimuli, supporting the potential of personalized sensory-based horticultural interventions.

2. Materials and Methods



Participants: 33 adults (27 women, 6 men; mean age 38.6 ± 11.9 years)



Psychological state grouping: Low_PSS (n=9), Mid_PSS (n=24), Good_PSQI (n=13), Bad_PSQI (n=20)



Stimuli: 24 tactile stimuli = 12 real vegetables + 12 artificial counterparts

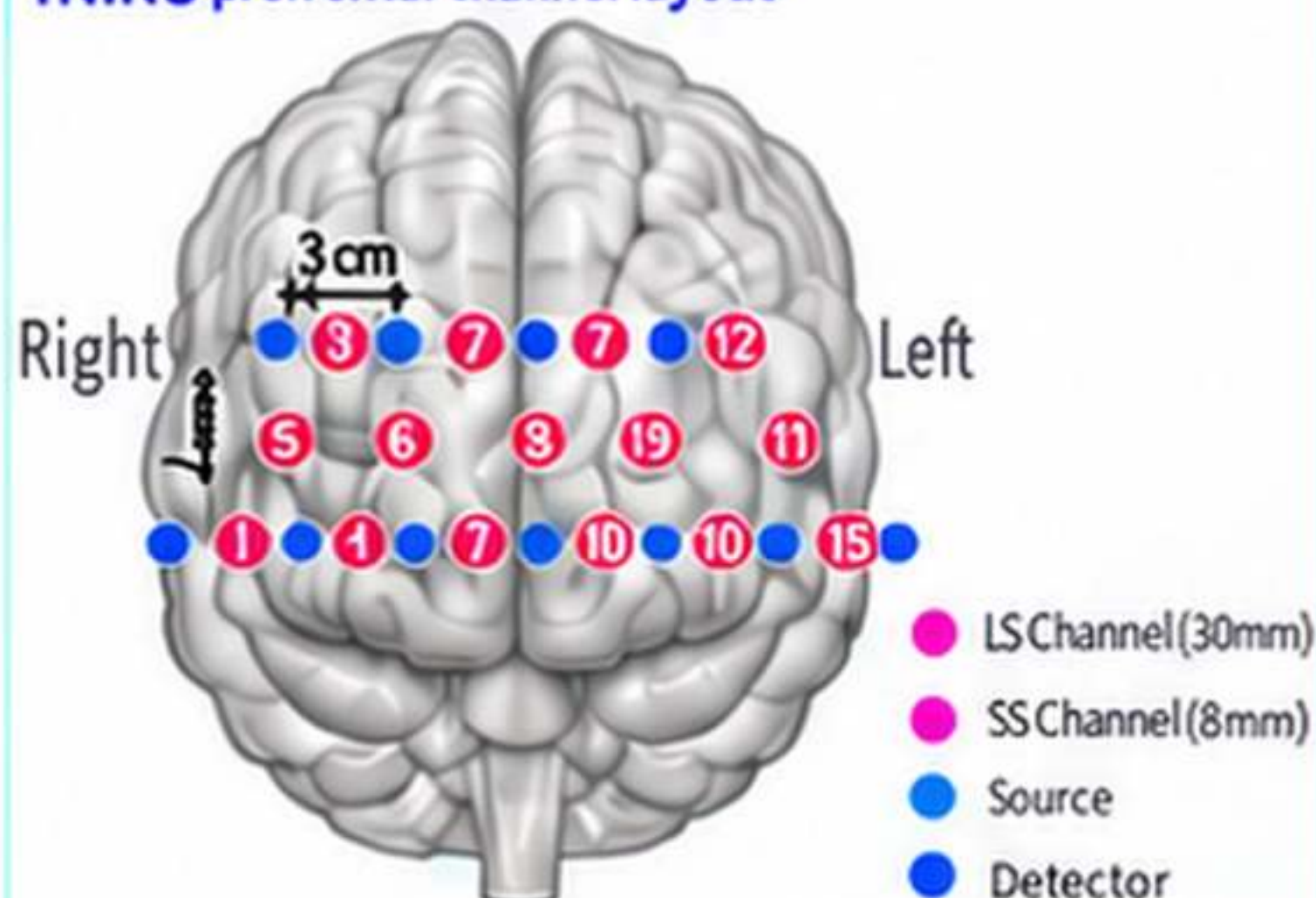


Procedure: 20s tactile exploration + 20 s rest; positive emotion rated on a 7-point Likert scale



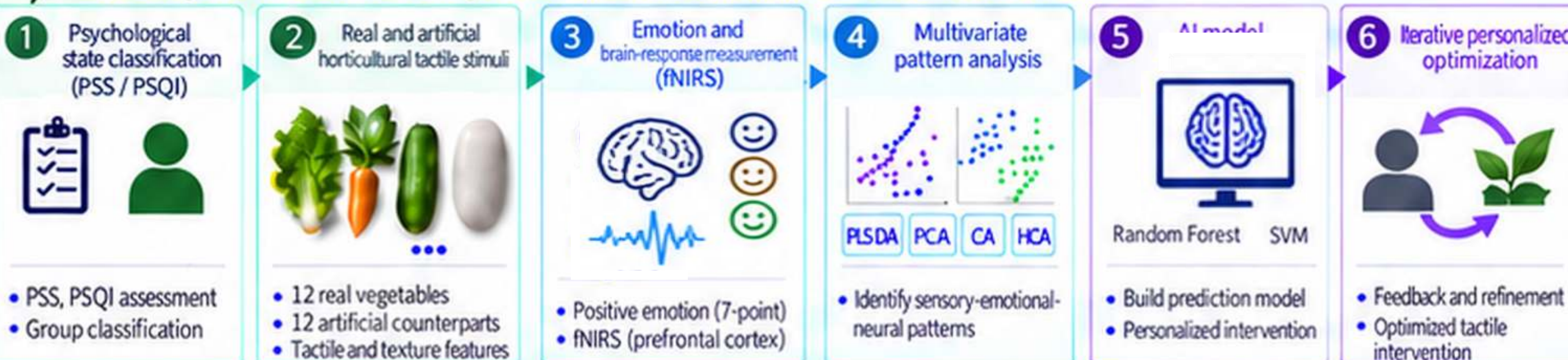
Neurophysiology: fNIRS (OBELAB NIRSIT LITE), 13 prefrontal channels analyzed

fNIRS prefrontal channel layout



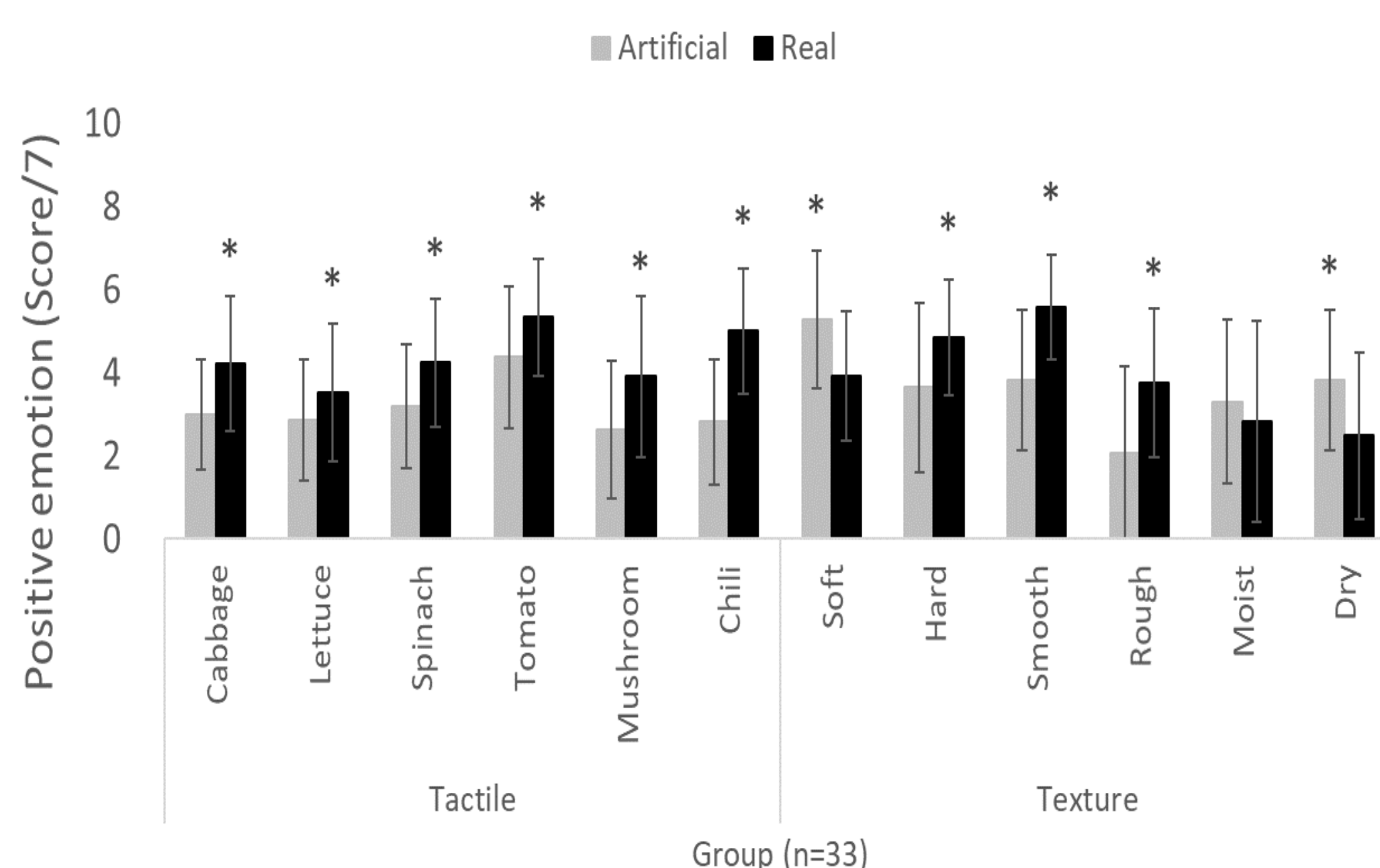
Statistics and modeling: paired t-test, PLS-DA, PCA, CA, HCA

Six-step framework for AI-personalized horticultural tactile intervention

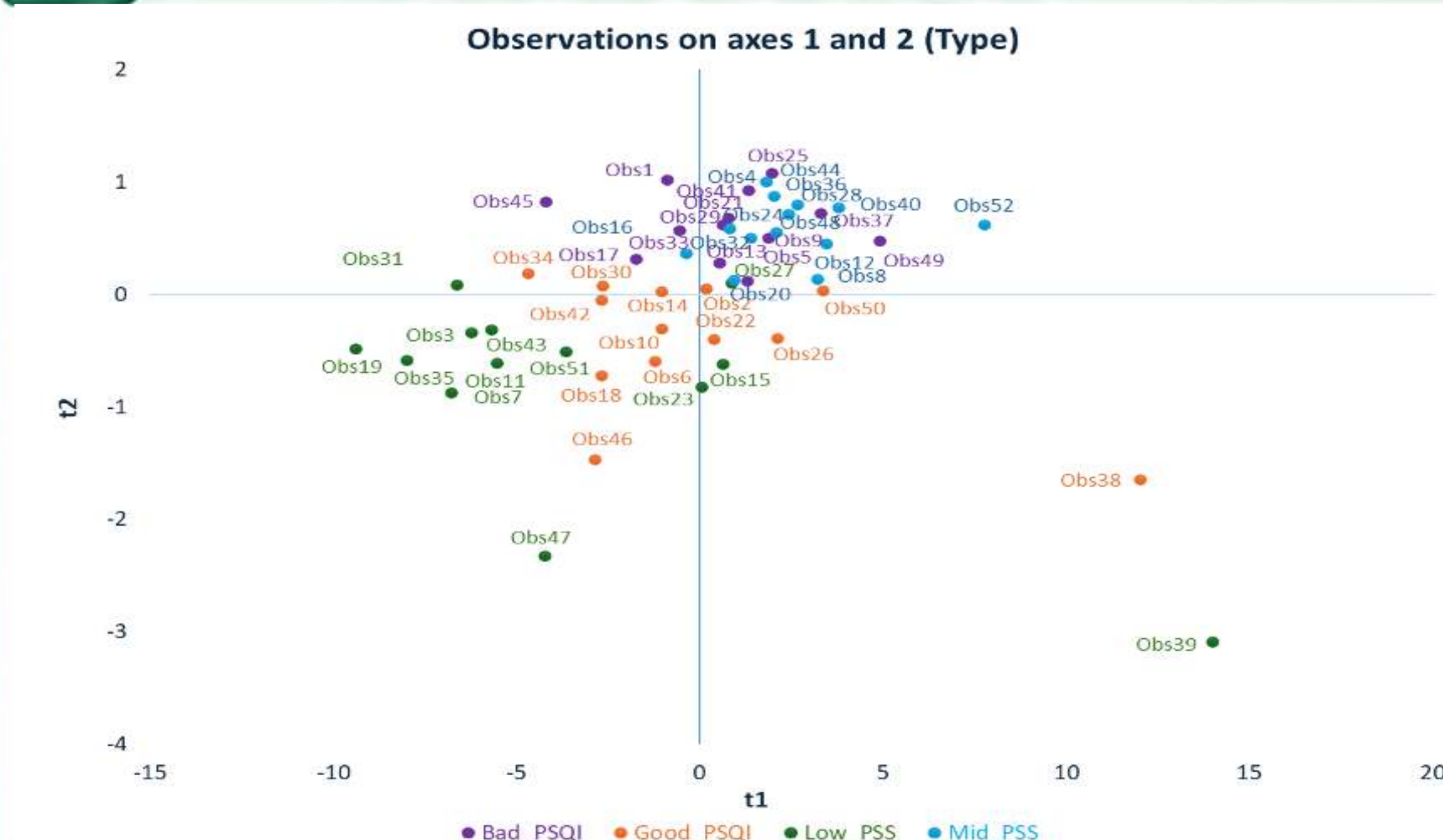


3. Results

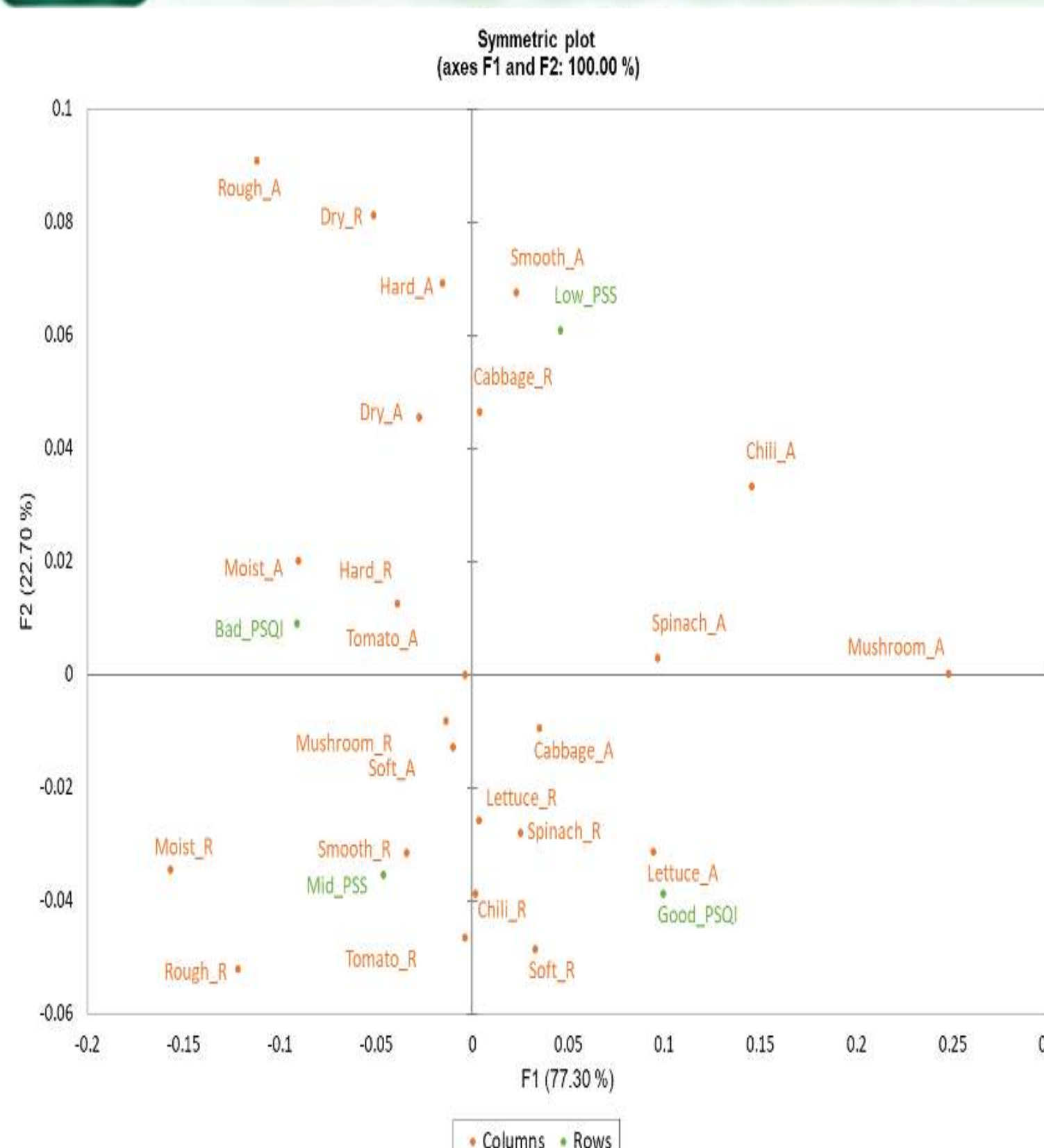
R1. Positive emotion scores for real vs artificial materials



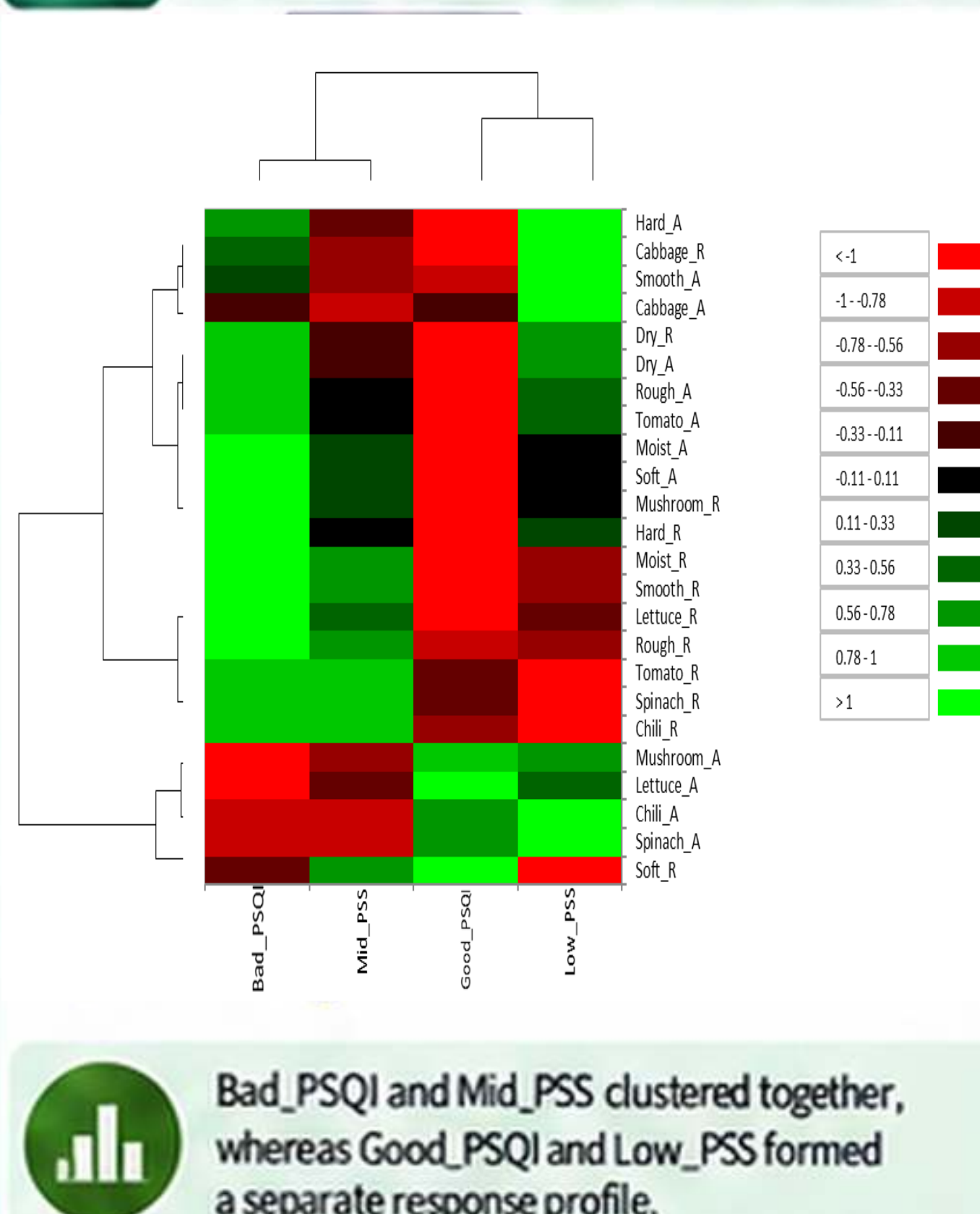
R2. PLS-DA reveals stress- and sleep-related discriminators



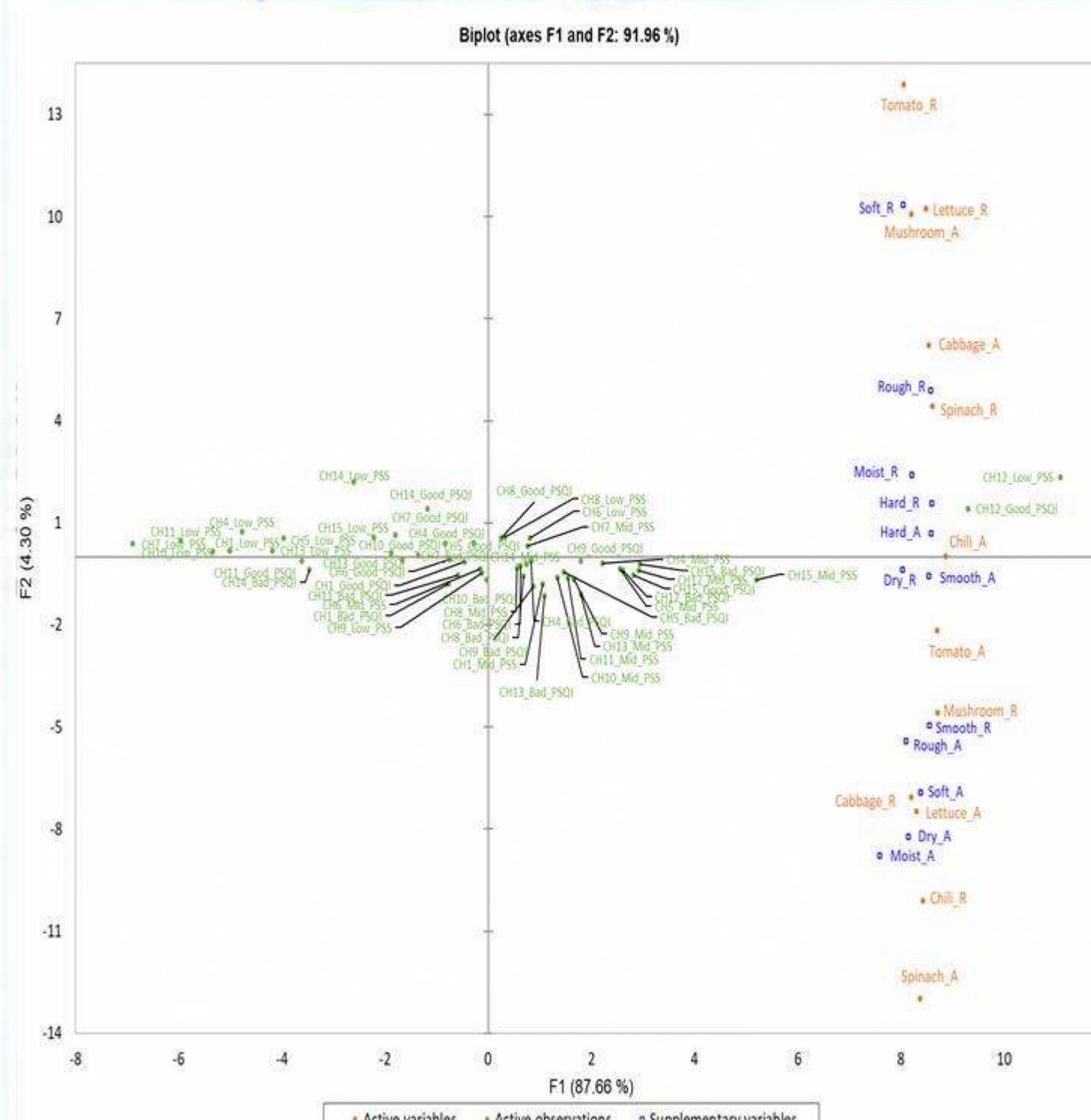
R3. Correspondence analysis identifies group-specific tactile preferences



R4. Hierarchical clustering supports distinct response profiles



R5. PCA suggests left prefrontal involvement in positive tactile response



4. Conclusion

- Real vegetable materials generally elicited stronger positive emotional responses than artificial counterparts, especially across tactile categories.
- Perceived stress and sleep quality were associated with distinct tactile response patterns, suggesting partly independent modulation pathways.
- Chili_R and Cabbage_R were key stress-related discriminators, whereas Tomato_R showed the strongest sleep-related contribution.
- CH12 and CH15 co-varied with positive tactile response patterns, suggesting possible involvement of left prefrontal regions.
- These findings provide preliminary, data-driven support for AI-personalized horticultural tactile intervention.

5. Acknowledgements

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6. References

- McGlone, F. & Krappelt, L. (2014). D. Horticulture and Affective touch.
- Treld, C. Cognitive science, 11, 176-182.
- Prehn, A. et al. (2020). NIRS in cognitive and emotional studies. *Neuroscience, Biobehavioral and Health*.
- V6, L. W. H. et al. (2012). Psychological and physiological responses to urban green. *Int. J. Environ. Res. Public Health*, 15, 67-06.
- Amur, Rev Clin Psychol. 14, 679-706.